



Ritik Singh

React / Full-Stack Engineer · DevOps · AI/RAG

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PROFESSIONAL SUMMARY

Full-Stack Engineer with ~2 years of hands-on experience building responsive, production-grade web applications. Specialize in React.js frontends, Node.js/Express backend APIs, and MongoDB data layers. Experienced in cloud deployment on AWS with Docker, Linux server management (NGINX, Apache), and Cloudflare DNS automation. Also built an AI-powered RAG system using embeddings and FAISS vector search. Passionate about clean architecture, real-time systems, and continuously adopting emerging technologies.

TECHNICAL SKILLS

Languages	JavaScript (ES6+), Python, C, HTML5, CSS3, Bash
Frontend	React.js, Responsive Design, REST API Integration, Socket.IO Client
Backend	Node.js, Express.js, Flask, REST APIs, JWT Auth, WebSocket
Databases	MongoDB, MySQL, MariaDB
DevOps & Cloud	AWS (EC2, S3), Docker, Terraform, Ansible
Servers & OS	Linux (Ubuntu, CentOS, RHEL), NGINX, Apache, Cloudflare
AI / RAG	RAG Pipeline, FAISS (Vector Search), Semantic Search, Embeddings, NLP Basics, OpenAI API
Tools	Git, GitHub, GitLab, Postman, VS Code

PROJECTS

Real-Time Chat Application	Role: React / Full-Stack Developer ● Live ■
<i>React.js · Node.js · Express.js · MongoDB · Socket.IO · JWT · Git</i>	
- Built full-stack MERN chat platform supporting real-time one-to-one messaging with instant delivery via Socket.IO/WebSocket. - Designed MongoDB schema for users and conversations; implemented JWT-based authentication and session management. - Developed responsive React.js frontend with online-status indicators, message history, and smooth UX. - Optimized WebSocket event handling for concurrent multi-user environments.	
iQuiz — Real-Time Quiz Platform (Kahoot-like)	Role: React / Full-Stack Developer ● Live ■
<i>React.js · Node.js · Express.js · MongoDB · Socket.IO · HTML · CSS</i>	
- Designed and built a multiplayer live-quiz platform with PIN-based session joining and real-time score updates. - Implemented full CRUD for quizzes and questions; built live-hosting feature generating unique PINs per session. - Managed real-time participant interactions and leaderboard updates using Socket.IO events. - Delivered seamless integration between React frontend and Node/Express backend.	
RAG-Based Intelligent Data Query System	Role: AI / Platform Engineer
<i>Python · FAISS · Pandas · OpenAI API · Sentence Transformers · NumPy</i>	
- Architected a RAG pipeline: document chunking → embedding generation → FAISS vector indexing → LLM response synthesis. - Integrated semantic search for efficient, context-aware retrieval over custom datasets of growing size. - Built hybrid retrieval combining vector search with structured querying (Pandas/SQL) for multi-condition aggregations. - Ensured responses are strictly grounded in provided data, reducing hallucinations in LLM output.	

Cloudflare DNS Automation System

Role: DevOps / Backend Developer

Node.js · AWS EC2 · Cloudflare API · Bash

- Built automated system that detects AWS EC2 public IP changes and propagates updates to Cloudflare DNS records instantly.
- Eliminated manual DNS management overhead; reduced DNS propagation lag on IP rotation.

Invoice PDF Generator System

Role: Backend Developer

Node.js · Express.js · PDF Library · REST API

- Developed a Node.js service that dynamically generates professional, branded PDF invoices from structured data.
- Exposed REST API endpoints for invoice creation and on-demand client download.

EXPERIENCE

React / Full-Stack Engineer (Self-Employed / Freelance)

2023 – Present

Jaipur, Rajasthan, India

- Designed and shipped multiple full-stack web applications covering frontend, backend API, database design, and deployment.
- Built real-time systems (chat & quiz platforms) using MERN stack and WebSocket/Socket.IO technologies.
- Deployed and managed applications on AWS EC2; configured NGINX reverse proxy, SSL, and Cloudflare DNS automation.
- Containerized services with Docker; automated deployment workflows using shell scripting and Bash.
- Developed an AI/RAG system demonstrating applied machine learning and FAISS vector search capabilities.

EDUCATION

BCA — Bachelor of Computer Applications

2023 – Present

GLA University, Mathura

12th Standard (Intermediate)

Passed 2023

UP Board

10th Standard (High School)

Passed 2021

UP Board

KEY HIGHLIGHTS

- 5+ end-to-end projects shipped independently — from architecture to deployment.
- Production experience with real-time systems handling concurrent users via WebSocket/Socket.IO.
- Hands-on AI/ML engineering: built a working RAG pipeline with FAISS vector search and LLM integration.
- Cloud & DevOps breadth: AWS EC2, Docker, Terraform, NGINX, Bash automation.
- Quick learner with a bias for action — comfortable navigating frontend, backend, infra, and AI layers.